

# Arthur Fournier

MACHINE LEARNING · ROBOTICS · PERCEPTION

Paris, France · Seeking an internship from April 2027

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## EDUCATION

### Mines Paris – PSL

DOUBLE DEGREE WITH ÉCOLE POLYTECHNIQUE · ROBOTICS TRACK

- Nonlinear control of manipulators and mobile robots, reinforcement learning for robotics, estimation and filtering, 3D vision.

### École polytechnique

MSc IN ENGINEERING (INGÉNIEUR POLYTECHNICIEN PROGRAM) · TOP 20% OF CLASS

- Machine learning, deep learning, reinforcement learning, generative AI, computer vision, statistics, optimization and control.

### Lycée Hoche

PREPARATORY CLASSES IN MATHEMATICS, PHYSICS AND COMPUTER SCIENCE (MPI)

## EXPERIENCE

### NUS & IPAL (CNRS)

RESEARCH INTERN, EVENT-BASED VISION FOR ASSISTIVE ROBOTS

- Cut the trajectory error of DEVO, the state-of-the-art learned event-camera odometry, by **41%** on EVIMO2 by designing a 1.0M-parameter causal ConvLSTM that down-weights independently moving objects inside patch selection and differentiable bundle adjustment.
- Showed that end-to-end joint training is key, capturing over 90% of a ground-truth mask's gain vs. 60% when trained separately, and that a label-free variant self-supervised by the bundle adjustment still cuts error by **37%**.
- Benchmarked 5 configurations on 6 datasets (EVIMO1/2, MVSEC, EDS, TUM-VIE, VECtor) on a Slurm GPU cluster, improving the baseline in 15 of 20 zero-shot evaluations, with a paper and open-source release in progress.

### Headmind Partners AI

AI ENGINEER INTERN

- Designed and deployed a multi-agent LLM pipeline where role-played HR, CTO and CEO agents assess each CV, then rank candidates.
- Delivered it as a finished internal tool (OpenAI API, Hugging Face embeddings, Streamlit, unit tests), used to screen 50+ CVs.

## PROJECTS

### Makeitalive: Animating a Single Landscape Photo

PYTORCH, DIFFUSERS

- With a teammate, iterated from an optical-flow U-Net to LoRA adapters (3.3M parameters) to full fine-tuning of Stable Video Diffusion's temporal layers on 2,374 drone clips (12 h on an A100), turning a single photo into a video with cinematic drone-like parallax.

### Reinforcement Learning for Block Blast

PYTORCH, GYMNASIUM, WEIGHTS & BIASES

- In a team of 4, increased average survival **4.7x** over a greedy baseline (209 vs. 44 moves over 250 games) by replacing DQN and Rainbow with an afterstate value network 63x smaller (42k parameters) using row and column detector convolutions.

### edge-mppi: Real-time GPU Model Predictive Control (in progress)

CUDA, C++, PYTHON, NVIDIA JETSON

- Building an MPPI controller around a hand-written fused CUDA kernel that keeps the vehicle state in registers for the whole horizon, targeting 8,192 rollouts in under 20 ms on a Jetson, with Nsight profiling and a benchmark against nav2\_mppi\_controller.

### Retrieval Assistant for Emergency Call Handlers

RAG, FAISS, LLAMA

- In a student team, built an on-premise RAG assistant over a 190-page medical regulation guide, with LLM page routing before FAISS retrieval and source-linked answers, tested in a SAMU call center after interviews and live-call observation with operators.

### X-Forum Careers Fair Platform

REACT, TYPESCRIPT, DJANGO, DOCKER, CI/CD

- Co-built the bilingual portal of École polytechnique's careers fair, serving **250 companies** and 3,000 students and processing 160 stand orders: one-time-code login, 9-step ordering wizard, DocuSign-signed purchase orders, sales-pipeline dashboard.

## SKILLS

|                  |                                                                                                            |
|------------------|------------------------------------------------------------------------------------------------------------|
| Programming      | Python, C++, C, CUDA, Java, SQL, OCaml, TypeScript, HTML/CSS                                               |
| Machine Learning | PyTorch, Hugging Face (Transformers, Diffusers), scikit-learn, XGBoost, pandas, FAISS                      |
| Systems & Tools  | Git, Docker, CI/CD, Slurm, Weights & Biases, React, Django, Streamlit                                      |
| Languages        | French (native), English (C1), Spanish (B2)                                                                |
| Leadership       | Gendarmerie officer cadet, aide to the commander of a 250-strong company (2023–24), X-Forum logistics team |
| Interests        | Basketball (university team), BMX racing (10 years, inter-regional), piano and guitar                      |